

7. GOVERNANCE & RELEASE GATE · STPA Hazard Analysis · Bumpless Human Override · Release Case ('LOOP-01' → 'REL-01')**1. SENSING**
Photons / Voltages to DMA · MIPI/SPI**2. PERCEPTION**
ViT Encoders · DINOv2 · 3D Tokens**3. MEMORY**
Temporal Belief · $SE(3)$ Trees · JEPAs**5. PLANNING (System 1.5 · MPU/NPU)**
Diffusion Policies · ACT Chunking (H -steps) · Jerk Continuity**4. REASONING (System 2 · MPU)**
Vision-Language Models · 3D Goals · Leases (t_{expire})

- - - - - THE PROPOSAL-PERMISSION PRIVILEGE BOUNDARY - - - - -

6. REFLEX (System 1 · MCU)
1 kHz Real-Time Loop · Control Barrier Functions ($h(x) \geq 0$) · Stopping Distance $d_{\text{stop}} \leq d_{\text{clearance}} \cdot \text{Veto}(u_t)$ **THE PHYSICAL WORLD ($W_t \rightarrow W_{t+1}$)** · Kinetic Momentum · Joule Heat · Friction · Collision Dynamics